

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [143, 195, 137, 181, 143, 156, 168, 142, 164, 170]

Tap amplitudes (pixels): [157.8131806916013, 174.72549899771354, 151.32745950421557, 168.34488409215172, 151.38031576133008, 153.44380078712857, 153.45683432157722, 136.4734406395618, 135.88230201170424, 118.0889495253472, 140.75865870347016]

GPT-5

MDS-UPDRS Item 3.4 Finger Tapping – Analysis

Given features are for a sequence of 10 consecutive taps (periods list has 10 intervals, implying 11 extrema detected but 10 inter-tap periods). We'll assess speed, rhythm interruptions/hesitations, and amplitude trend/decrements.

1) Tap count and rhythm/interruptions

- Number of taps reported: 11, but 10 periods provided → consistent with 10 taps analyzed.
- Periods (frames): [143, 195, 137, 181, 143, 156, 168, 142, 164, 170]
 - Mean period $\approx$ 160 frames with variability.
 - Notable longer intervals: 195 and 181 frames (2 conspicuous slowdowns). Others are relatively clustered (137–170).
 - Clear “interruptions/hesitations” count: approximately 2 distinct longer hesitations (near taps 2 and 4). No evidence of >5 interruptions or a clear long arrest/freeze.

Interpretation: One to two interruptions/hesitations fits the “Slight” criterion (1a). Total does not reach 3–5 (which would be Mild, 2a), nor >5 or a freeze (Moderate, 3a).

2) Speed

- With the periods mostly in a moderate range and only two notably longer ones, overall slowing appears slight rather than mild/moderate.
- This aligns with “Slight slowing” (1b), not clearly “Mild slowing”.

3) Amplitude and decrement pattern

- Amplitudes (pixels): [157.8, 174.7, 151.3, 168.3, 151.4, 153.4, 153.5, 136.5, 135.9, 118.1, 140.8]
 - Early taps are larger (150–175). From tap 8 onward there’s a decline: 136.5 → 135.9 → 118.1, then a partial rebound to 140.8.
 - Decrement is not immediate after the 1st tap; it becomes more evident toward the end (notably taps 9–10). That pattern corresponds to:
 - “Slight” (1c): amplitude decrements near the end of the 10 taps.
 - It does not meet “Mild” (2c: decrements midway) nor “Moderate” (3c: decrements starting after the 1st tap).

Overall synthesis

- Interruptions/hesitations: about two → Slight.
- Speed: slight slowing → Slight.
- Amplitude: decrement near the end → Slight.

No criterion strongly supports Mild (2) or Moderate (3), and performance is clearly above Severe (4).

Final score: 1
